GEOMETRY · UNIT 1: FOUNDATIONS OF GEOMETRY

Treasure Map Navigator

Use distance, midpoint, and angle relationships to find buried treasure.

CP2

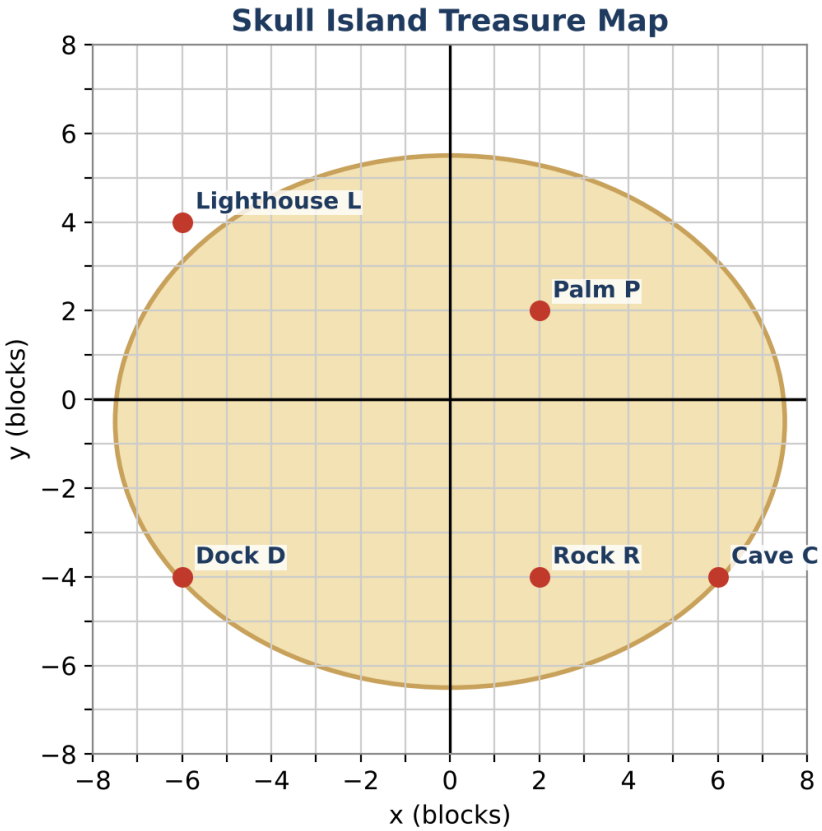
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

A pirate's map of Skull Island has turned up in the school library. The landmarks sit on a coordinate grid where each unit is one block.

Each clue uses a geometry idea: distance, midpoint, segment addition, or angle relationships. Solve the clues to find the treasure, then make a map of your own.



Learning Goals

- I can find distances and midpoints on the coordinate plane.
- I can use segment addition and angle addition to solve for unknowns.
- I can identify vertical angles and linear pairs.

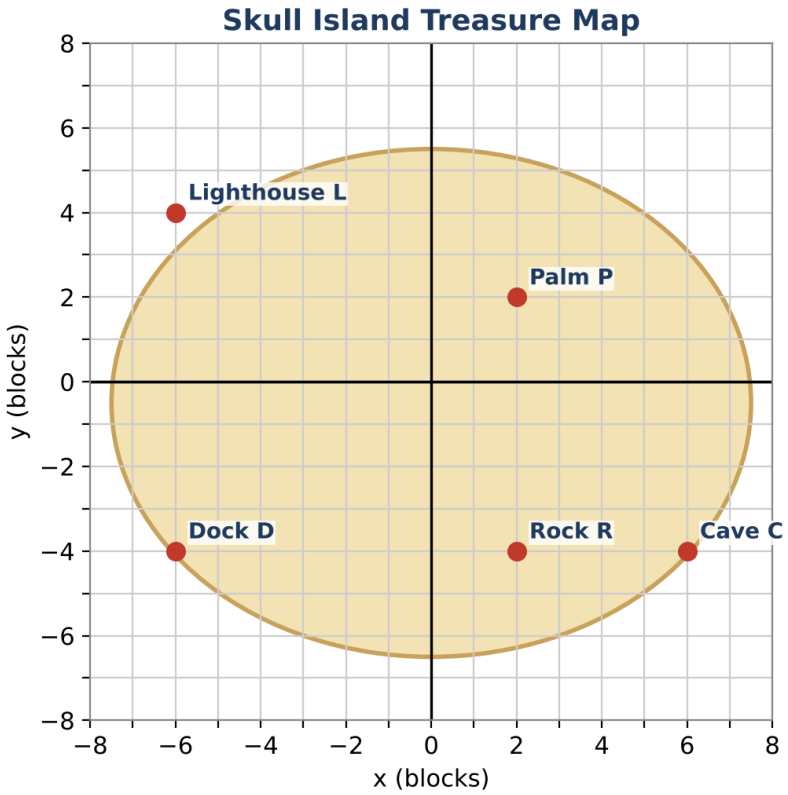
Project Timeline

When	What you will do
Day 1	Distance & Midpoint Clues
Day 2	Segment & Angle Clues
Day 3	Design Your Own Map
Final	Treasure Map & Clue Set + reflection

Part 1: Distance & Midpoint Clues

Day 1 · Use the map and the coordinate table.

1. Clue 1: How far is it from the Dock to the Cave? (They are on the same horizontal line.)



Location	Coordinates
Dock D	(−6, −4)
Palm P	(2, 2)
Cave C	(6, −4)
Lighthouse L	(−6, 4)
Rock R	(2, −4)

2. Clue 2: How far is it from the Dock to the Palm? Draw a right triangle on the grid and use the Pythagorean Theorem.

Starter: *legs: 8 and 6*

3. Clue 3: The first chest is buried at the midpoint of the Dock and the Palm. Find its coordinates.

   Starter: *Average the x-coordinates, then average the y-coordinates.*

4. Clue 4: The second chest is at the midpoint of the Palm and the Cave. Find its coordinates.

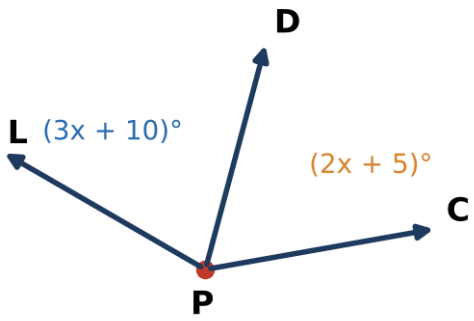
Part 2: Segment & Angle Clues

Day 2 · Use algebra to unlock each clue.

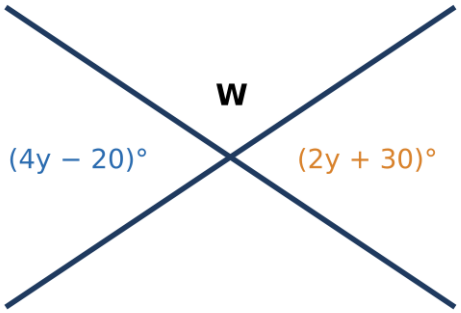
5. The Dock, the Rock, and the Cave are on one straight path. DR = 2x + 2 and RC = x + 1, and DC = 12. Find x, DR, and RC.

6. At the Palm, ray PD splits ∠LPC. m∠LPD = (3x + 10)°, m∠DPC = (2x + 5)°, and m∠LPC = 95°. Find x and both angle measures.

Angles at the Palm (not to scale)



Paths crossing at the Well



7. Two paths cross at the Well and form vertical angles of (4y − 20)° and (2y + 30)°. Find y and the angle measure.

8. What is the measure of an angle that forms a linear pair with the angle from #7?

Part 3: Design Your Own Map

Day 3 · Make a treasure map on graph paper.

9. Place at least four landmarks on a coordinate grid and list their coordinates.

10. Write a midpoint clue and a distance clue that lead to your treasure. Solve them in an answer key.

11. Write one angle clue (angle addition or vertical angles) with an answer key.

Final Product: Treasure Map & Clue Set

Your final product must include:

- A labeled treasure map on a coordinate grid
- At least three clues (midpoint, distance, angle)
- An answer key with all work
- A short pirate story

Reflection

R1. Why does the midpoint formula average the coordinates?

R2. How did the Pythagorean Theorem help you find a diagonal distance?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Original clues correctly use distance, midpoint, and angle relationships and lead to one location. Goes beyond the minimum with a creative, realistic choice.	Original clues correctly use distance, midpoint, and angle relationships and lead to one location.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Map and clue set show labeled coordinates, every clue, and a complete answer key. Polished, organized, and visually engaging.	Map and clue set show labeled coordinates, every clue, and a complete answer key. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.